

Домашнее задание № 10

1. Запустим 4 экземпляра ClickHouse (2 шарда × 2 реплики)

 docker-compose.yml – Блокнот

Файл Правка Формат Вид Справка

version: '3.8'

services:

zookeeper:

image: zookeeper:3.8

container_name: zookeeper

hostname: zookeeper

restart: unless-stopped

ports:

- "2181:2181"

environment:

ZOO_MY_ID: 1

ZOO_SERVERS: server.1=zookeeper:2888:3888;2181

networks:

- clickhouse-network

Шард 1, Реплика 1

clickhouse-shard1-replica1:

image: clickhouse/clickhouse-server:latest

container_name: clickhouse-shard1-replica1

hostname: clickhouse-shard1-replica1

restart: unless-stopped

ports:

- "8123:8123"

- "9000:9000"

- "9009:9009"

environment:

- CLICKHOUSE_DB=\${CLICKHOUSE_DB:-default}

- CLICKHOUSE_USER=\${CLICKHOUSE_USER:-default}

- CLICKHOUSE_PASSWORD=\${CLICKHOUSE_PASSWORD}

- CLICKHOUSE_LOG_LEVEL=information

volumes:

- ./config/macros-shard1-replica1.xml:/etc/clickhouse-server/config.d/macros.xml

- ./config/keeper.xml:/etc/clickhouse-server/config.d/keeper.xml

- ./config/clusters.xml:/etc/clickhouse-server/config.d/clusters.xml

- clickhouse-data-shard1-replica1:/var/lib/clickhouse

- clickhouse-logs-shard1-replica1:/var/log/clickhouse-server

depends_on:

- zookeeper

networks:

- clickhouse-network

Шард 1, Реплика 2

clickhouse-shard1-replica2:

image: clickhouse/clickhouse-server:latest

container_name: clickhouse-shard1-replica2

hostname: clickhouse-shard1-replica2

restart: unless-stopped

ports:

- "8124:8123"

- "9001:9000"

- "9010:9009"

environment:

- CLICKHOUSE_DB=\${CLICKHOUSE_DB:-default}

- CLICKHOUSE_USER=\${CLICKHOUSE_USER:-default}

- CLICKHOUSE_PASSWORD=\${CLICKHOUSE_PASSWORD}

- CLICKHOUSE_LOG_LEVEL=information

volumes:

- ./config/macros-shard1-replica2.xml:/etc/clickhouse-server/config.d/macros.xml

- ./config/keeper.xml:/etc/clickhouse-server/config.d/keeper.xml

- ./config/clusters.xml:/etc/clickhouse-server/config.d/clusters.xml

- clickhouse-data-shard1-replica2:/var/lib/clickhouse

- clickhouse-logs-shard1-replica2:/var/log/clickhouse-server

depends_on:

- zookeeper

networks:

- clickhouse-network

```
# Шард 2, Реплика 1
clickhouse-shard2-replica1:
  image: clickhouse/clickhouse-server:latest
  container_name: clickhouse-shard2-replica1
  hostname: clickhouse-shard2-replica1
  restart: unless-stopped
  ports:
    - "8125:8123"
    - "9002:9000"
    - "9011:9009"
  environment:
    - CLICKHOUSE_DB=${CLICKHOUSE_DB:-default}
    - CLICKHOUSE_USER=${CLICKHOUSE_USER:-default}
    - CLICKHOUSE_PASSWORD=${CLICKHOUSE_PASSWORD}
    - CLICKHOUSE_LOG_LEVEL=information
  volumes:
    - ./config/macros-shard2-replica1.xml:/etc/clickhouse-server/config.d/macros.xml
    - ./config/keeper.xml:/etc/clickhouse-server/config.d/keeper.xml
    - ./config/clusters.xml:/etc/clickhouse-server/config.d/clusters.xml
    - clickhouse-data-shard2-replica1:/var/lib/clickhouse
    - clickhouse-logs-shard2-replica1:/var/log/clickhouse-server
  depends_on:
    - zookeeper
  networks:
    - clickhouse-network

# Шард 2, Реплика 2
clickhouse-shard2-replica2:
  image: clickhouse/clickhouse-server:latest
  container_name: clickhouse-shard2-replica2
  hostname: clickhouse-shard2-replica2
  restart: unless-stopped
  ports:
    - "8126:8123"
    - "9003:9000"
    - "9012:9009"
  environment:
    - CLICKHOUSE_DB=${CLICKHOUSE_DB:-default}
    - CLICKHOUSE_USER=${CLICKHOUSE_USER:-default}
    - CLICKHOUSE_PASSWORD=${CLICKHOUSE_PASSWORD}
    - CLICKHOUSE_LOG_LEVEL=information
```

...

2. Создадим 2 топологии:

- Топология 1: 2 шарда × 1 реплика
- Топология 2: 1 шард × 2 реплики

Конфигурационные файлы:



keeper.xml – Блокнот

Файл Правка Формат Вид Справка

```
<clickhouse>
  <zookeeper>
    <node>
      <host>zookeeper</host>
      <port>2181</port>
    </node>
  </zookeeper>
</clickhouse>
```

Макросы для каждого инстанса:



macros-shard1-replica1.xml – Блокнот

Файл Правка Формат Вид Справка

```
<clickhouse>
  <macros>
    <shard>1</shard>
    <replica>replica-1</replica>
  </macros>
</clickhouse>
```



macros-shard1-replica2.xml – Блокнот

Файл Правка Формат Вид Справка

```
<clickhouse>
  <macros>
    <shard>1</shard>
    <replica>replica-2</replica>
  </macros>
</clickhouse>
```

 macros-shard2-replica1.xml – Блокнот	 macros-shard2-replica2.xml – Блокнот
Файл Правка Формат Вид Справка	Файл Правка Формат Вид Справка
<pre><clickhouse> <macros> <shard>2</shard> <replica>replica-1</replica> </macros> </clickhouse></pre>	<pre><clickhouse> <macros> <shard>2</shard> <replica>replica-2</replica> </macros> </clickhouse></pre>

Описание топологий:

 clusters.xml – Блокнот
Файл Правка Формат Вид Справка

```
<clickhouse>
  <remote_servers>
    <!-- Топология 1: 2 шарда, 1 реплика -->
    <cluster_2shards_1replica>
      <shard>
        <replica>
          <host>clickhouse-shard1-replica1</host>
          <port>9000</port>
          <user>default</user>
          <password from_env="CLICKHOUSE_PASSWORD" />
        </replica>
      </shard>
      <shard>
        <replica>
          <host>clickhouse-shard2-replica1</host>
          <port>9000</port>
          <user>default</user>
          <password from_env="CLICKHOUSE_PASSWORD" />
        </replica>
      </shard>
    </cluster_2shards_1replica>

    <!-- Топология 2: 1 шард, 2 реплики -->
    <cluster_1shard_2replicas>
      <shard>
        <replica>
          <host>clickhouse-shard1-replica1</host>
          <port>9000</port>
          <user>default</user>
          <password from_env="CLICKHOUSE_PASSWORD" />
        </replica>
        <replica>
          <host>clickhouse-shard1-replica2</host>
          <port>9000</port>
          <user>default</user>
          <password from_env="CLICKHOUSE_PASSWORD" />
        </replica>
      </shard>
    </cluster_1shard_2replicas>
```

```

<!-- Топология 3: 2 шарда, 2 реплики (полная) -->
<cluster_2shards_2replicas>
  <shard>
    <replica>
      <host>clickhouse-shard1-replica1</host>
      <port>9000</port>
      <user>default</user>
      <password from_env="CLICKHOUSE_PASSWORD" />
    </replica>
    <replica>
      <host>clickhouse-shard1-replica2</host>
      <port>9000</port>
      <user>default</user>
      <password from_env="CLICKHOUSE_PASSWORD" />
    </replica>
  </shard>
  <shard>
    <replica>
      <host>clickhouse-shard2-replica1</host>
      <port>9000</port>
      <user>default</user>
      <password from_env="CLICKHOUSE_PASSWORD" />
    </replica>
    <replica>
      <host>clickhouse-shard2-replica2</host>
      <port>9000</port>
      <user>default</user>
      <password from_env="CLICKHOUSE_PASSWORD" />
    </replica>
  </shard>
</cluster_2shards_2replicas>
</remote_servers>
</clickhouse>

```

3. Создание локальных и распределенных таблиц

```

-- На clickhouse-shard1-replica1 (порт 8123)
CREATE TABLE one_local ON CLUSTER cluster_2shards_2replicas (
  dummy UInt8
)
ENGINE = ReplicatedMergeTree('/clickhouse/tables/{shard}/one_local', '{replica}')
ORDER BY dummy;

-- Вставляем тестовые данные на каждый шард
INSERT INTO one_local VALUES (1),(2),(3);

```

Создаем DISTRIBUTED таблицы для каждой топологии

```

-- Топология 1: 2 шарда, 1 реплика
CREATE TABLE one_distributed_2shards_1replica
ENGINE = Distributed('cluster_2shards_1replica', 'default', 'one_local', rand())
AS one_local;

-- Топология 2: 1 шард, 2 реплики
CREATE TABLE one_distributed_1shard_2replicas
ENGINE = Distributed('cluster_1shard_2replicas', 'default', 'one_local', rand())
AS one_local;

-- Топология 3: 2 шарда, 2 реплики
CREATE TABLE one_distributed_2shards_2replicas
ENGINE = Distributed('cluster_2shards_2replicas', 'default', 'one_local', rand())
AS one_local;

```

4. Проверка DISTRIBUTED таблиц

-- Топология 1: 2 шарда, 1 реплика

```
SELECT
    hostname() as local_host,
    _shard_num,
    dummy,
    count() as count
FROM one_distributed_2shards_1replica
GROUP BY local_host, _shard_num, dummy
ORDER BY _shard_num;
```

Результат 1 X

SELECT hostname() as local_host, _shard_num, dummy, count() as count Введите SQL выражение чтобы отфильтровать результаты

	A-Z local_host	123 _shard_num	123 dummy	123 count
1	clickhouse-shard1-replica1	1	3	1
2	clickhouse-shard1-replica1	1	1	1
3	clickhouse-shard1-replica1	1	2	1
4	clickhouse-shard2-replica1	2	2	1
5	clickhouse-shard2-replica1	2	3	1
6	clickhouse-shard2-replica1	2	1	1

-- Топология 2: 1 шард, 2 реплики

```
SELECT
    hostname() as local_host,
    _shard_num,
    dummy,
    count() as count
FROM one_distributed_1shard_2replicas
GROUP BY local_host, _shard_num, dummy
ORDER BY local_host;
```

Результат 1 X

SELECT hostname() as local_host, _shard_num, dummy, count() as count Введите SQL выражение чтобы отфильтровать результаты

	A-Z local_host	123 _shard_num	123 dummy	123 count
1	clickhouse-shard1-replica1	1	1	1
2	clickhouse-shard1-replica1	1	2	1
3	clickhouse-shard1-replica1	1	3	1

SELECT hostname() as local_host, _shard_num, dummy, count() as count Введите SQL выражение чтобы отфильтровать результаты

	A-Z local_host	123 _shard_num	123 dummy	123 count
1	clickhouse-shard1-replica2	1	1	1
2	clickhouse-shard1-replica2	1	2	1
3	clickhouse-shard1-replica2	1	3	1

-- Топология 3: 2 шарда, 2 реплики
SELECT
 hostname() as local_host,
 _shard_num,
 dummy,
 count() as count
FROM one_distributed_2shards_2replicas
GROUP BY local_host, _shard_num, dummy
ORDER BY _shard_num, local_host;

результат 1

SELECT hostname() as local_host, _shard_num, dummy, count() as count

	A-Z local_host	123 _shard_num	123 dummy	123 count
1	clickhouse-shard1-replica2	1	1	1
2	clickhouse-shard1-replica2	1	3	1
3	clickhouse-shard1-replica2	1	2	1
4	clickhouse-shard2-replica1	2	2	1
5	clickhouse-shard2-replica1	2	3	1
6	clickhouse-shard2-replica1	2	1	1

SELECT hostname() as local_host, _shard_num, dummy, count() as count

	A-Z local_host	123 _shard_num	123 dummy	123 count
1	clickhouse-shard1-replica1	1	3	1
2	clickhouse-shard1-replica1	1	1	1
3	clickhouse-shard1-replica1	1	2	1
4	clickhouse-shard2-replica2	2	1	1
5	clickhouse-shard2-replica2	2	3	1
6	clickhouse-shard2-replica2	2	2	1

Проверка системных кластеров:

SELECT
 cluster,
 shard_num,
 replica_num,
 host_name
FROM system.clusters
ORDER BY cluster, shard_num, replica_num;

clusters 1

SELECT cluster, shard_num, replica_num, host_name

	A-Z cluster	123 shard_num	123 replica_num	A-Z host_name
1	cluster_1shard_2replicas	1	1	clickhouse-shard1-replica1
2	cluster_1shard_2replicas	1	2	clickhouse-shard1-replica2
3	cluster_2shards_1replica	1	1	clickhouse-shard1-replica1
4	cluster_2shards_1replica	2	1	clickhouse-shard2-replica1
5	cluster_2shards_2replicas	1	1	clickhouse-shard1-replica1
6	cluster_2shards_2replicas	1	2	clickhouse-shard1-replica2
7	cluster_2shards_2replicas	2	1	clickhouse-shard2-replica1
8	cluster_2shards_2replicas	2	2	clickhouse-shard2-replica2

```
SHOW CREATE TABLE one_distributed_2shards_1replica;

CREATE TABLE default.one_distributed_2shards_1replica
(
  `dummy` UInt8
)
ENGINE = Distributed('cluster_2shards_1replica', 'default', 'one_local', rand())

SHOW CREATE TABLE one_distributed_1shard_2replicas;

CREATE TABLE default.one_distributed_1shard_2replicas
(
  `dummy` UInt8
)
ENGINE = Distributed('cluster_1shard_2replicas', 'default', 'one_local', rand())

SHOW CREATE TABLE one_distributed_2shards_2replicas;

CREATE TABLE default.one_distributed_2shards_2replicas
(
  `dummy` UInt8
)
ENGINE = Distributed('cluster_2shards_2replicas', 'default', 'one_local', rand())
```
